IMO 2026 — Problem 2

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-23

Source: `results/gpt-5.6-sol-pro/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

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Approaches tried

- **Angle parameters and polar coordinates.** The midpoint conditions and sine rule reduce the problem to a trigonometric factorization.
- **Circle intersections with AB, AC.** If the circumcircle of AKL meets the oriented lines AB, AC again at D, E, then $OM = ON$ is exactly $AB \cdot AD - AC \cdot AE = (AB^2 - AC^2)/2$.
- **Numerical/symbolic checking.** `code/explore_trig.py` checked compatible samples; `code/symbolic_poly.py` exposed the exact harmonic factorization and its factor $\sin(x-y)$.
- **Harmonic factorization (successful).** The target residual is exactly $\sin(x-y)$ times the compatibility residual. A complete proof is in `lemmas/trigonometric-factorization.md`.

Current best

The problem is solved. With suitable angle parameters, the two midpoint conditions determine AK/AB and AL/AC. The two remaining angle conditions yield a compatibility equation. An oriented-circle calculation turns $OM = ON$ into a trigonometric target, and the factorization lemma shows that its residual is $\sin(x - y)$ times the already-zero compatibility residual.

Full proof

Let

$$p = \angle KBA = \angle ACL, \quad q = \angle LBK = \angle LNC, \quad r = \angle LCK = \angle BMK,$$

and put

$$x = \angle BAK, \qquad y = \angle CAL, \qquad \delta = \angle KAL.$$

Because K lies inside $\angle LBA$, the ray BK lies between BA and BL; because L lies inside $\angle ACK$, the ray CL lies between CA and CK. We first justify the resulting order of the rays at A. Write the positive normalized affine coordinates of K, L relative to A, B, C as

$$K = (k_A, k_B, k_C), \qquad L = (l_A, l_B, l_C).$$

At B, the first betweenness says $k_C/k_A < l_C/l_A$; at C, the second says $l_B/l_A < k_B/k_A$. Therefore

$$\frac{k_B}{l_B} > \frac{k_A}{l_A} > \frac{k_C}{l_C}.$$

Thus $k_C/k_B < l_C/l_B$. Indeed, a ray from A through a point with affine coordinates (u_A, u_B, u_C) has direction $u_B\overrightarrow{AB} + u_C\overrightarrow{AC}$; consequently this last inequality says precisely that the ray AK lies between AB and AL. Since both points are inside ABC, the ray order is consequently AB, AK, AL, AC. Hence

$$\angle BAC = x + \delta + y. \qquad (0)$$

Put $c = AB$ and $b = AC$.

Applying the sine rule in ABK and BMK, and using $BM = c/2$, gives

$$BK = \frac{c \sin x}{\sin(p + x)} = \frac{c \sin r}{2 \sin(p + r)}.$$

Consequently

$$\cot x = \cot p + 2 \cot r, \qquad AK = cP, \qquad P := \frac{\sin p}{\sin(p + x)}. \qquad (1)$$

Similarly, the sine rule in ACL and CNL, using $CN = b/2$, gives

$$\cot y = \cot p + 2 \cot q, \qquad AL = bQ, \qquad Q := \frac{\sin p}{\sin(p + y)}. \qquad (2)$$

Now apply the sine rule in ACK and ABL. From (0),

$$\angle CAK = \delta + y, \qquad \angle BAL = x + \delta.$$

Also, the angle assumptions give

$$\angle ACK = p + r, \qquad \angle ABL = p + q.$$

Thus, on setting

$$U = p + r + \delta + y, \qquad V = p + q + \delta + x,$$

we obtain

$$\frac{c}{b}P = \frac{\sin(p+r)}{\sin U}, \qquad \frac{b}{c}Q = \frac{\sin(p+q)}{\sin V}. \qquad (3)$$

Multiplying these equalities yields

$$PQ \sin U \sin V = \sin(p+r)\sin(p+q). \qquad (4)$$

Let the circle through A, K, L meet the oriented lines AB, AC again at D, E, respectively, and write $AD = d$, $AE = e$ as directed lengths. Take A as origin, and let $\mathbf{e}_B, \mathbf{e}_C$ be the unit vectors along AB, AC. If $\mathbf{w} = 2\overrightarrow{AO}$, the circle has equation

$$|\mathbf{z}|^2 = \mathbf{w} \cdot \mathbf{z}.$$

It follows that

$$d = \mathbf{w} \cdot \mathbf{e}_B, \qquad e = \mathbf{w} \cdot \mathbf{e}_C. \qquad (5)$$

Since $\overrightarrow{AM} = (c/2)\mathbf{e}_B$ and $\overrightarrow{AN} = (b/2)\mathbf{e}_C$, we have

$$OM^2 - ON^2 = \frac{c^2 - b^2}{4} - \frac{cd - be}{2}.$$

Therefore $OM = ON$ is equivalent to

$$cd - be = \frac{c^2 - b^2}{2}. \qquad (6)$$

Let $\mathbf{u}_K, \mathbf{u}_L$ be the unit vectors along AK, AL. The circle equation at K, L says

$$\mathbf{w} \cdot \mathbf{u}_K = cP, \qquad \mathbf{w} \cdot \mathbf{u}_L = bQ.$$

The angles from AB to AK, from AK to AL, and from AL to AC are x, δ, y, respectively. Solving these two linear equations and then taking the projections of $\mathbf{w}$ on AB, AC gives

$$d = \frac{cP\sin(\delta + x) - bQ\sin x}{\sin\delta}, \qquad e = \frac{-cP\sin y + bQ\sin(\delta + y)}{\sin\delta}. \qquad (7)$$

From the first relation in (1), a direct sine expansion gives

$$P\sin(\delta + x) - \frac{1}{2}\sin\delta = \frac{\sin p}{2\sin(p+r)}\sin(\delta + r). \qquad (8)$$

Analogously, (2) gives

$$Q\sin(\delta + y) - \frac{1}{2}\sin\delta = \frac{\sin p}{2\sin(p+q)}\sin(\delta + q). \qquad (9)$$

Substitution of (7)–(9) shows that (6) is equivalent to

$$\frac{c}{b}\frac{\sin p\sin(\delta+r)}{2\sin(p+r)}-\frac{b}{c}\frac{\sin p\sin(\delta+q)}{2\sin(p+q)}+P\sin y-Q\sin x=0. \qquad (10)$$

Using (3), equation (10) is equivalent to $T=0$, where

$$\begin{aligned}T&=Q\sin p\sin(\delta+r)\sin V-P\sin p\sin(\delta+q)\sin U\\&\quad+2(P\sin y-Q\sin x)\sin(p+r)\sin(p+q). \qquad (11)\end{aligned}$$

We use the following trigonometric identity, proved below:

$$T=\sin(x-y)\big(PQ\sin U\sin V-\sin(p+r)\sin(p+q)\big). \qquad (12)$$

Indeed, the relations $\cot x=\cot p+2\cot r$ and $\cot y=\cot p+2\cot q$ imply

$$P=\frac{\sin r}{2\sin(r-x)}, \qquad Q=\frac{\sin q}{2\sin(q-y)}, \qquad (13)$$

and

$$\frac{\sin(p+r)}{\sin p}=\frac{\sin(r-x)}{\sin x}, \qquad \frac{\sin(p+q)}{\sin p}=\frac{\sin(q-y)}{\sin y}. \qquad (14)$$

From (13), putting the expression in brackets below over a common denominator, and using the sine subtraction formula, we obtain

$$\begin{aligned}&[2(P\sin y-Q\sin x)+\sin(x-y)]\sin(p+r)\sin(p+q)\\&=\sin^2 p\sin(q+x-r-y). \qquad (15)\end{aligned}$$

For completeness, the numerator used here is

$$\begin{aligned}&\sin r\sin y\sin(q-y)-\sin q\sin x\sin(r-x)\\&\quad+\sin(x-y)\sin(r-x)\sin(q-y)\\&=\sin x\sin y\sin(q+x-r-y),\end{aligned}$$

and (14) then gives (15).

It remains to verify (12). Put

$$W=2\delta+p+q+r, \qquad D=q+x-r-y.$$

After applying $2\sin s\sin t=\cos(s-t)-\cos(s+t)$, the part of twice the difference between the two sides of (12) that depends on δ is

$$-Q\sin p\cos(W+x)+P\sin p\cos(W+y)+\sin(x-y)PQ\cos(W+p+x+y). \qquad (16)$$

Divide (16) by PQ and use $\sin p/P=\sin(p+x)$ and $\sin p/Q=\sin(p+y)$. The result is

$$-\sin(p+x)\cos(W+x)+\sin(p+y)\cos(W+y)+\sin(x-y)\cos(W+p+x+y),$$

which is zero: applying $2\sin a\cos b=\sin(a+b)+\sin(a-b)$ makes its six terms cancel in pairs.

It remains to compare the constant parts. We first prove

$$\begin{aligned} &Q \sin p \cos(r - p - q - x) - P \sin p \cos(q - p - r - y) \\ &\quad - \sin(x - y)PQ \cos D = -2 \sin^2 p \sin D. \end{aligned} \tag{17}$$

After division by PQ, this is equivalent to

$$\begin{aligned} &\sin(p + x) \cos(p + y + D) - \sin(p + y) \cos(D - p - x) \\ &\quad - \sin(x - y) \cos D = -2 \sin(p + x) \sin(p + y) \sin D. \end{aligned} \tag{18}$$

Here we used $r - p - q - x = -(p + y + D)$ and $q - p - r - y = D - (p + x)$. To prove (18), expand the two cosines of sums and also $\sin(x - y) = \sin((p + x) - (p + y))$. All terms containing $\cos D$ cancel, while the two remaining terms are both $-\sin(p + x) \sin(p + y) \sin D$.

In twice the difference between the sides of (12), the complete constant part is the left side of (17) plus

$$[4(P \sin y - Q \sin x) + 2 \sin(x - y)] \sin(p + r) \sin(p + q).$$

By (15), this latter expression is $2 \sin^2 p \sin D$, which cancels the right side of (17). Thus both the variable and constant parts vanish, proving (12).

By (4), the right-hand side of (12) is zero. Hence $T = 0$, so (10), (6), and therefore $OM = ON$ follow.